

Important Questions

DBMS

- Examine the use of stored procedures and triggers in PL/SQL.
- Analyze and construct a PL/SQL program that accepts two numbers and displays their sum. Incorporate exception handling to manage invalid input
- Explain Two-Phase Locking (2PL) protocol and how it ensures serializability.
- Determine what a deadlock is in a database system. Explain the four necessary conditions required for a deadlock to occur. How can deadlocks be prevented in a database system ?
- Examine the purpose of "BEFORE" and "AFTER" triggers in PL/SQL. Provide examples of when each would be used.
- Analyze and construct a PL/SQL program that accepts two numbers and displays their sum. Incorporate exception handling to manage invalid input
- Explain the differences between Two-Phase Locking (2PL) and Optimistic Concurrency Control in transaction management.
- Determine what a deadlock is in a database system. Explain the four necessary conditions required for a deadlock to occur. How can deadlocks be prevented in a database system ?

NoSQL

- Evaluate the use of Redis in real-time applications. Compare it with MongoDB in terms of performance and storage.
- Implement a basic caching layer using Redis for an API response in a Node.js environment.
- Evaluate the process of designing a simple MongoDB-based application. Justify your approach to data modeling and reporting with suitable strategies.
- Design a simple blog project using MongoDB. Implement document structure, indexing, and basic aggregation.
- Evaluate Redis' role as an in-memory data store compared to MongoDB's document-oriented database. When is Redis a better option for storage, and when should MongoDB be preferred?
- Implement a basic caching layer using Redis for an API response in a Node.js environment.
- Imagine you are developing a live chat application, and you need to manage user sessions, store messages in a queue for real-time processing, and cache the most recent chat history for quick retrieval.
 - 1) Write the Redis command to store a session for a user with a 30-minute expiration time.
 - 2) Write the Redis command to cache the last 10 messages of a chat.
- Design a simple blog project using MongoDB. Implement document structure, indexing, and basic aggregation.

